



**UNIVERSITI SAINS MALAYSIA**

School of Electrical and Electronics Engineering  
Universiti Sains Malaysia, Engineering Campus

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**EEM 441**

Control & Instrumentation Laboratory

## **Experiment 4**

**INTERFACING FPGA TO DIP SWITCH AND SEVEN SEGMENT**

Date : \_\_\_\_\_

Group No. : \_\_\_\_\_

Group Members : \_\_\_\_\_ Matrix # \_\_\_\_\_

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# Experiment 4

## Interfacing FPGA to DIP Switch and Seven Segment

### Objectives

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1. To use Verilog code as a design entry method in Altera Quartus II Software.
2. To interface a DIP switch and a seven-segment display to the FPGA.

### Equipment / Software Required

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1. Altera Quartus II software
  - version 9/10/12 for Cyclone II
  - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Function generator

### Introduction

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Seven-segment displays are widely used to show numbers. The DE2 board provides eight seven-segment displays, 18 DIP switches, four push-button switches, and 16 LEDs. These components are hardwired on the board and are each assigned to specific FPGA pins — refer to the DE2 user manual for the full pin map.

### Procedure

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#### Activity 1 — 16-bit binary counter with frequency divider

*Objective: design a 16-bit binary counter driven through a frequency divider.*

1. Open Quartus II and write the frequency-divider module in Listing 4.1 (seven), which lowers the incoming clock frequency (Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N)).
2. Create a reusable symbol for seven.
3. Write the 4-bit counter module in Listing 4.2 (1ed).
4. Create a reusable symbol for 1ed.
5. Draw the schematic in Figure 4.1, connecting the seven block's most significant output bit q[16] to the clock input of the 1ed block.
6. Assign pin numbers as shown in Table 4.1 (refer to the DE2 user manual).

7. Use a function generator to supply the clock signal, using 1 MHz as the clock input.
8. Determine the effective input frequency reaching the led module.
9. Discuss your result based on the observed LED output display.

seven.v — frequency divider

```

1 module seven(q, clk);
2   output [15:0] q;
3   input clk;
4   reg [15:0] q;
5
6   always @(posedge clk)
7   begin
8     q = (q + 1) % 65536;
9   end
10 endmodule

```

Listing 4.1: seven.v — 16-bit frequency divider.

led.v — 4-bit counter

```

1 module led(x, clk);
2   input clk;
3   output [3:0] x;
4   reg [3:0] x;
5
6   always @(posedge clk)
7   begin
8     x = (x + 1) % 16;
9   end
10 endmodule

```

Listing 4.2: led.v — 4-bit binary counter.

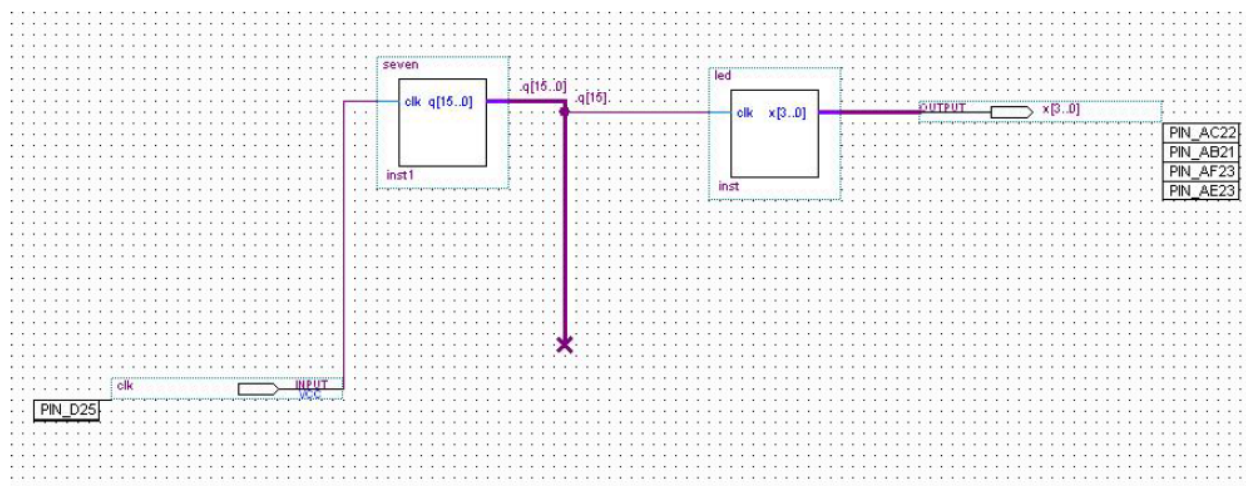


Figure 4.1: Schematic diagram for the 16-bit binary counter with frequency divider.

Table 4.1: Pin assignment for the 4-bit binary counter.

| # | Node name | Direction | Pin location |
|---|-----------|-----------|--------------|
| 1 | clk       | Input     |              |
| 2 | X[3]      | Output    |              |
| 3 | X[2]      | Output    |              |
| 4 | X[1]      | Output    |              |
| 5 | X[0]      | Output    |              |

### Activity 2 — 0–99 counter on seven-segment display

*Objectives: (1) design a counter that counts 0 to 99; (2) display the result on the seven-segment displays.*

1. Using a DIP switch as the trigger input, write Verilog code for a counter that counts from 0 to 99 once the switch is triggered.
2. Decode and display the result (tens and units digits) on two seven-segment displays.
3. Print and submit the Verilog code and simulation result.

### Activity 3 — Running light

*Objective: design a running-light display on the Altera DE2 board.*

1. Using a DIP switch as the input, write Verilog code to implement a running-light (chasing LED) pattern.
2. Use three DIP switches to select between three different running-light patterns.
3. Print and submit the Verilog code and simulation result.